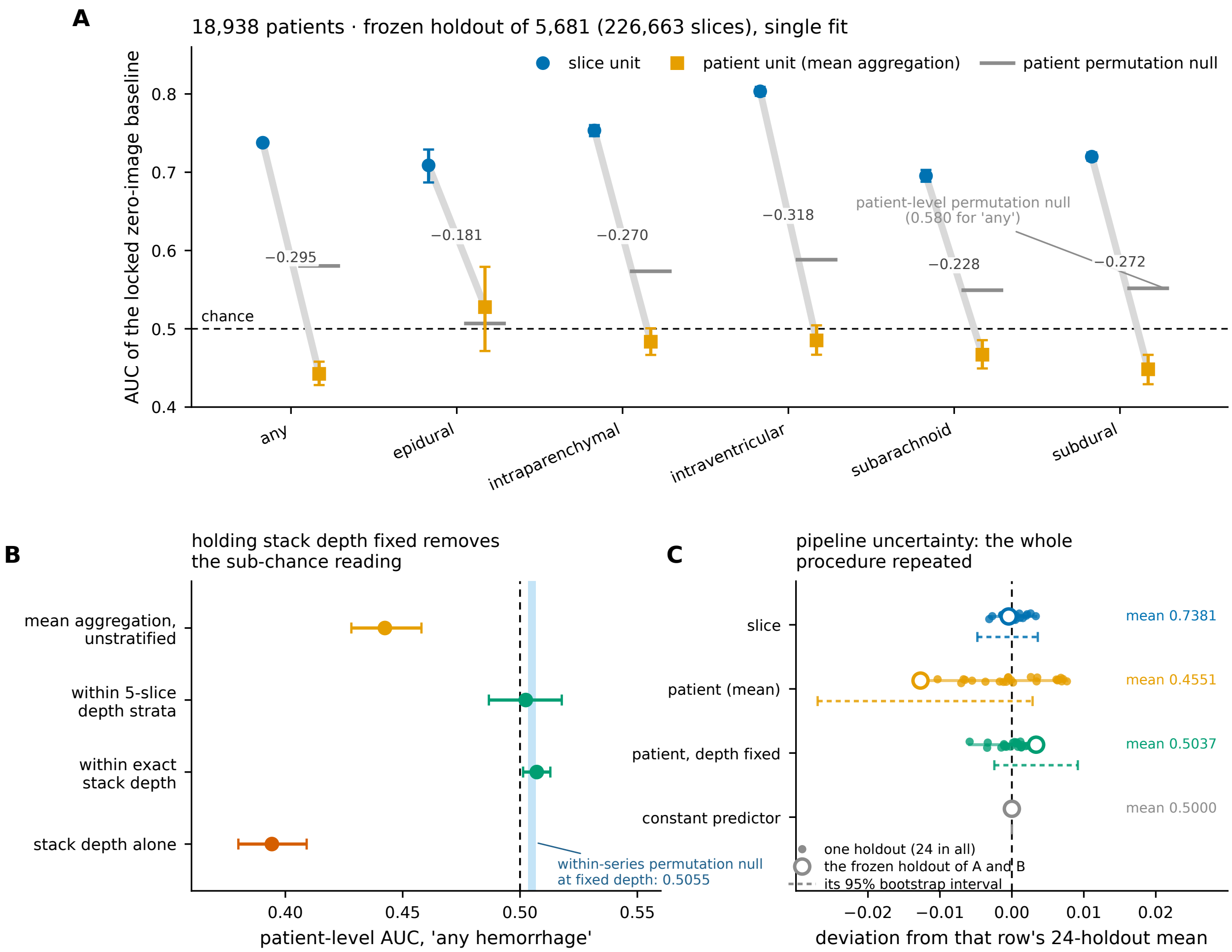

Figure 1. One locked pixel-blind score vector on the RSNA 2019 Intracranial Hemorrhage benchmark, read on a frozen patient-disjoint holdout (13,257 training patients; 5,681 held-out patients, 226,663 held-out slices; single fit, no pooling). **(A)** Slice-level area under the receiver operating characteristic curve (AUC) joined to patient-level AUC after mean aggregation for each of the six official labels, with 95% patient-clustered bootstrap intervals (2000 replicates) and the within-series label permutation null at the patient unit. Nothing changes between the two readings of a pair except the unit at which the ranking is performed. Three patient-level intervals lie wholly below chance and the other three cross it; epidural, on 108 positive patients, is the only point estimate above chance. The patient-level null lies above chance because permuting within a series preserves stack depth. **(B)** The mechanism, for the "any hemorrhage" label: the same patient-level score read unstratified, within exact stack depth, and within 5-slice depth strata, with stack depth read alone for comparison. Holding depth fixed removes the sub-chance reading; the depth-conditioned value of 0.5071 (95% CI: 0.5013, 0.5129) excludes 0.500 but sits 0.0016 above the within-series permutation null computed at fixed depth, 0.5055 (range, 0.5035 to 0.5068), shown as the shaded band, which its own interval spans. **(C)** The same four quantities over the 24 independently drawn holdouts of the identical design that furnish the primary estimate. Points are the individual holdouts, the bar is the family range, and the open marker is the pre-registered holdout of panels A and B, which sits at the edge of the family on the patient-level and depth-conditioned readings; that is why the family mean and not this draw is reported as primary. The constant predictor is exactly 0.500 in every holdout, which is the property the superseded pooled out-of-fold estimator did not have.

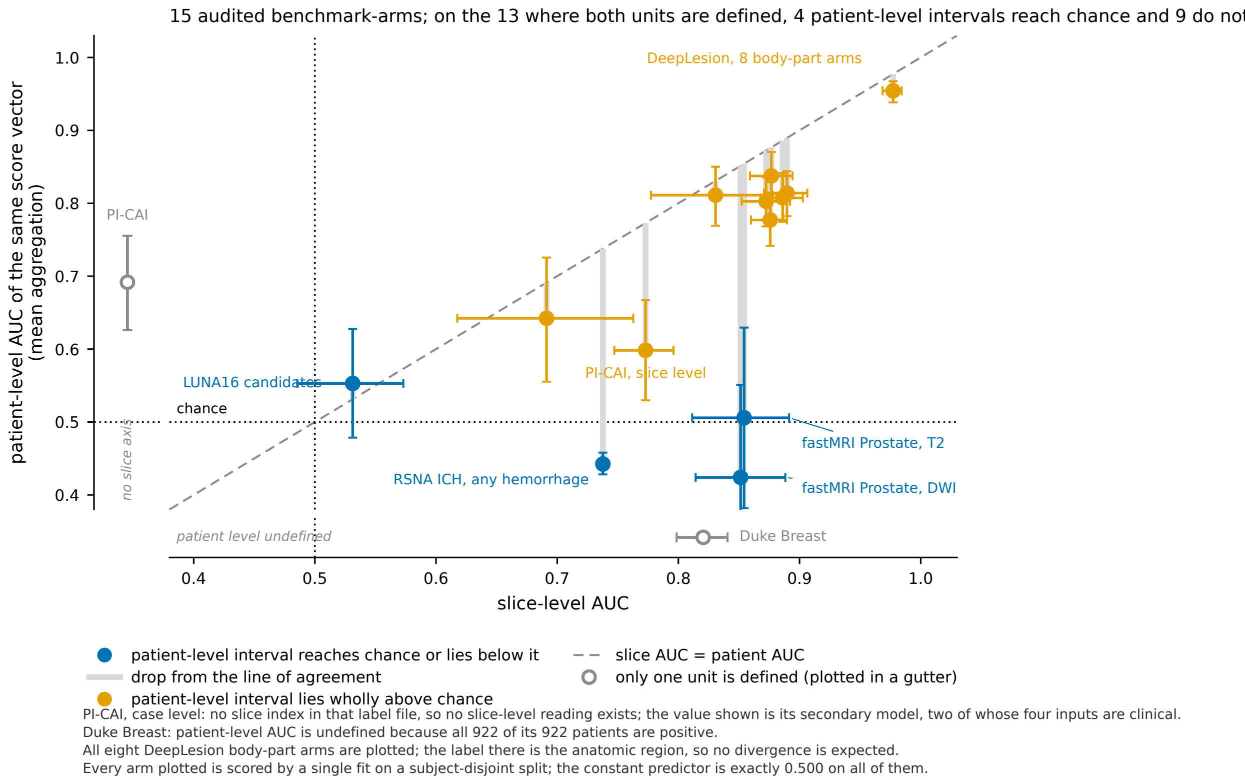

Figure 2. The same pixel-blind score vector read at two units, for every audited benchmark-arm. Each point is one arm: its slice-level area under the receiver operating characteristic curve (AUC) against the patient-level AUC of the identical score vector after mean aggregation, with 95% subject-clustered bootstrap intervals on both axes (replicate counts in Table 4). The dashed diagonal is the line on which the two units agree, and the grey stem below each point is the distance that arm falls from it. All eight DeepLesion body-part arms are plotted, not one of them: they are the arms on which no divergence occurs, and that is expected rather than anomalous, because the label there is the anatomic region and relative slice position predicts it at either unit. Values are for the locked 20-bin positional baseline except PI-CAI, whose positional baseline is exactly 0.500 on the public marksheet, which carries no slice index, and whose secondary tree over that file's four non-image columns—two of them clinical, age and prostate-specific antigen level—is drawn instead. That benchmark's slice-level arm, read from its 1,295 lesion delineation volumes under the same locked protocol, is plotted separately and at both units. Every arm plotted is scored by a single fit on a subject-disjoint split and its constant predictor is exactly 0.500. The two arms defined at only one unit are drawn in the margins rather than in the field, so that a value that does not exist cannot be read as a low one: PI-CAI's label file carries no slice index, and Duke Breast's patient-level AUC is undefined because all 922 of its 922 patients are positive.

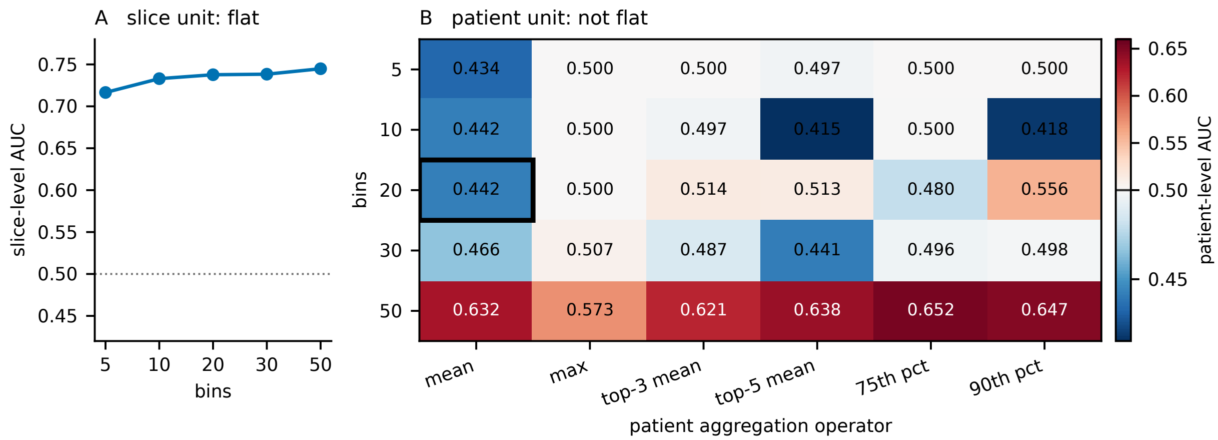

Figure 3. How the two units respond to the two analysis choices the positional baseline requires. **(A)** Slice-level AUC against the number of position bins: 0.717, 0.733, 0.738, 0.738 and 0.745 over 5, 10, 20, 30 and 50 bins, a range of 0.028. **(B)** Patient-level AUC over the same bin counts crossed with six aggregation operators, on the same frozen holdout of 5,681 patients and the same single fit. The boxed cell, 20 bins with mean aggregation, is the pre-specified analysis reported throughout; every other cell is a sensitivity analysis and none was used to choose it. Colour is centred on 0.500. The patient reading spans 0.415 to 0.652, a range of 0.237, against 0.028 at the slice unit for the identical fits. Two features carry the paper's caveat rather than its claim: at 50 bins every operator reads above chance, because a bin then holds about one slice and the aggregate begins to track stack depth directly; and the operators that read near exactly 0.500 are degenerate, taking only a handful of distinct values across all 5,681 patients. The slice-level reading is nearly invariant to both choices. The patient-level reading is not, and any patient-level statement made here is a statement about mean aggregation at 20 bins.